



IAEA-TERC-2025-01

Worldwide Proficiency Test Exercise on the determination of anthropogenic and natural radionuclides in water, soil, vegetation and simulated contaminated surface samples.

Important - before you start to analyse these samples.

Please kindly check the inventory of the package for completeness (Table 1) and ensure that the samples are not damaged.

Should any items be missing or broken, please send an email to:
Proficiency-Tests.Contact-Point@iaea.org.

Instructions to Participants

1. Description of the samples

The package contains 5 samples, 2 QC samples and one blank sample, as detailed in Table 1. The sample set contains:

- five PT samples, for which results shall be reported and will be assessed,
- two quality control samples to support participants in measurements,
- and one blank sample.

The content of radionuclides in the samples is less than the exemption level, both in terms of activity concentrations and absolute activity values.

Table 1. Packing list.

Sample Code	Sample Matrix	Weight (approx.) / amount	Sample description
1	Spiked drinking water	500 g	Water spiked with naturally occurring and anthropogenic radionuclides
2	Natural mineral water	500 g	Water sample containing environmentally occurring radionuclides
3	Soil	200 g	Soil containing environmentally occurring radionuclides
4	Vegetation	100 g	Vegetation containing environmentally occurring radionuclides
5	Simulated contaminated surface sample printed on canvas substrate	1 pc	Printed contaminated surface containing a mixture of Ba-133 and Sr-90
(QC-1)	Simulated contaminated surface sample printed on canvas substrate	1 pc	Printed quality control sample containing Ba-133
(QC-2)	Simulated contaminated surface sample printed on canvas substrate	1 pc	Printed quality control sample containing Sr-90
	Blank surface sample (canvas substrate)	1 pc	Blank canvas sample

1.1. Sample 1 – Drinking water (spiked)

Drinking water was gravimetrically spiked with a known amount of a standard solution, prepared from individual solutions with a certified activity concentration of radionuclides. The batch of water was acidified to $\text{pH} < 2$ ($\sim 0.05\text{M HNO}_3$) for stabilization, and thoroughly mixed prior to bottling. Each bottle contains approximately 500 g of water.

The target values for activity concentration and corresponding uncertainties were assigned by formulation, and control measurements were performed by gamma-ray spectrometry, alpha-particle spectrometry and liquid scintillation counting.

1.2. Sample 2 – Natural mineral water (non-spiked)

Natural mineral water was acidified to $\text{pH} < 2$ ($\sim 0.05\text{M HNO}_3$) for stabilization, and thoroughly mixed prior to bottling. Each bottle contains approximately 500 g of water.

The water contains naturally occurring radionuclides at low levels. Specifically, uranium is present at a mass fraction between 0.1 ng/g and 15 ng/g.

The water contains elevated mass concentrations of certain inorganic ions compared to typical drinking water, including:

Calcium (Ca^{2+}): between 100 and 500 mg/L

Magnesium (Mg^{2+}): between 20 and 200 mg/L

Sulphate (SO_4^{2-}): between 100 and 500 mg/L

Sodium (Na^+): between 10 and 100 mg/L

The determination of target values and associated uncertainties of the radionuclides has been carried-out using alpha-particle spectrometry and ICP-MS.

1.3. Sample 3 - Soil

Approximately 200 g of soil is provided for gamma-ray spectrometry measurement. The bottled samples were sterilized (25 kGy gamma dose) and the related irradiation certificate is available on the PT website. It contains environmentally occurring radionuclides.

The determination of activity concentration target values and associated uncertainties of the radionuclides has been carried-out using high precision semiconductor gamma-ray spectrometry measurements in compliance with ISO/IEC 17025. The sample was characterized for gamma-emitting radionuclides only.

1.4. Sample 4 - Vegetation

Approximately 100 g of vegetation sample is provided for gamma-ray spectrometry measurement. The bottled samples were sterilized (25 kGy gamma dose) and the related certificate is available on the PT website. It contains environmentally occurring radionuclides.

The determination of target values and associated uncertainties of the radionuclides has been carried-out using high precision semiconductor gamma-ray spectrometry measurements in compliance with ISO/IEC 17025. The samples are characterized for gamma-emitting radionuclides only.

1.5. Sample 5 - Simulated contaminated surface sample (white envelope)

A set of printed canvas circles for contamination surface monitoring have been included in this proficiency test (Figure 1). They were prepared using an in-house printing technique. One blank sample of the same size, outlined with black ink (non-spiked) is also added for correction (Blank).

The simulated contaminated surface sample (purple) is printed with a combination of magenta ink (Sr-90) and blue ink (Ba-133), and the activity of both radionuclides should be reported (Sample 05).

The simulated contaminated surface sample (blue) is a QC sample (QC-1) and was spiked with a known activity of Ba-133 (see Table 3). The simulated contaminated surface sample (magenta) is a QC sample (QC-2) and was spiked with a known activity of Sr-90 (see Table 3).

The provided QC samples are intended to support the validation and/or calibration of measurement instruments to facilitate the determination of an unknown Sample 05.

The material used as printing substrate was Canvas Matt Fine Art, 380 M variant.

Table 3. Radionuclide content in simulated contaminated surface QC samples.

Sample	Sr-90 Bq/sample	Ba-133 Bq/sample
QC – 1	-	$7.36 \pm 0.25^*$
QC – 2	$4.39 \pm 0.08^*$	-

* Uncertainty is expressed as a combined standard uncertainty with coverage factor $k=1$

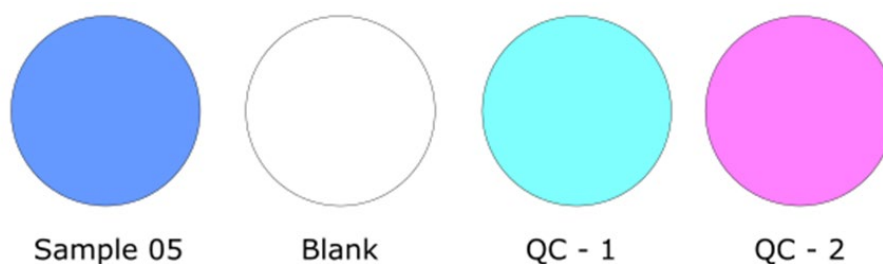


Figure 1. Simulated contaminated surface samples

2. Test sample handling

Please take care to avoid any cross contamination by ensuring all standard laboratory practices are applied and personal protective equipment is used. The activity of the samples is below the exemption level, however when radiochemistry is used to determine the activity concentration, care should be taken to avoid contamination of the laboratory.

2.1. Samples 1 and 2 - Water samples:

Thoroughly mix the sample by shaking before taking aliquots for measurement. It is recommended that any subsamples for analysis are measured by weight.

2.2. Samples 3 Soil and 4 Vegetation:

Thoroughly mix the sample before transferring the contents to your standard sample counting container for gamma-ray spectrometry. It is recommended that any subsamples for analysis are measured by weight. The minimum sample intake recommended for gamma-ray spectrometry measurements for soil and vegetation are 40 g and 30 g, respectively.

For soil and vegetation samples, dry content should be determined by drying several aliquots (a few grams each) of the sample immediately after opening the bottle. The recommended procedure for

drying is exposure to 105°C (soil sample) and 80°C (vegetation sample) in a drying oven for at least 4 hours (or to a constant mass).

2.3. Sample 5 and QC samples of simulated contaminated surfaces:

Only the area within the black outline has been printed with ink on the contaminated surface sample (purple/blue/magenta). The blank sample was printed with the black outline only (no radioactivity). The radionuclides are distributed evenly across the printed area and the stability and homogeneity of the samples have been confirmed.

To ensure the integrity of the samples, the printed surface should be kept dry and handled wearing gloves.

3. Measurements

3.1. Choice of method/procedure

Participants may use any method of their choice and the use of any relevant analytical techniques is acceptable.

The participating laboratory shall perform the same number of replicate determinations and statistical data treatment as for routine samples.

3.2. Requested results per sample

Samples 1, 3, 4 and 5 are intended for gamma-ray spectrometry measurements. Samples 1, 2 and 5 are intended for radiochemistry followed by alpha-particle spectrometry and/or liquid scintillation counting. Sample 2 provides the opportunity for measurement of uranium by ICP-MS as well as gross-alpha and gross-beta measurements. Table 4 provides additional details.

Table 4. Overview of requested results per sample.

Sample Code	Analytes	Reporting units
Sample 1 Water	Gamma-emitting radionuclides	Bq/kg
	^3H and ^{90}Sr and ^{210}Po (by radiochemistry)	Bq/kg
Sample 2 Water	^{234}U , ^{235}U , ^{238}U , ^{226}Ra , ^{228}Ra (by radiochemistry)	Bq/kg
	Gross alpha and gross beta	Bq/kg
	^{235}U , ^{238}U and total U (by ICP-MS)	ng/g
Sample 3 Soil	Gamma-emitting radionuclides	Bq/kg (d.m.)*
Sample 4 Vegetation	Gamma-emitting radionuclides	Bq/kg (d.m.)*
Sample 5 Canvas	^{133}Ba and ^{90}Sr	Bq/sample

* All results for samples 03 and 04 shall be reported based on dry mass.

The reference date for decay correction for all samples is January 1, 2025.

The decay correction is **not applicable** for the gross alpha and beta measurements.

4. Reporting requirements

The closing date of the reporting is October 1, 2025.

The determined activity concentration and its combined standard uncertainty ($k=1$) must be expressed as **activity concentration in Bq/kg** for sample 1 – drinking water, sample 2 - natural mineral water, sample 3 - soil and sample 4 – vegetation. Results for the latter two solid samples need to be corrected for dry mass content and reported in Bq/kg dry mass.

Important exception: Results for mass fractions of **uranium** isotopes in sample 2 natural mineral water, determined by ICP-MS, must be reported as **mass fractions in ng/g** with combined uncertainty of measurement in the same unit.

The measurement results for sample 5, simulated contaminated surface sample, must be reported as activity (Bq/sample) $\pm$ uncertainty (Bq/sample).

It is not mandatory to report measurement results for all samples and/or for all requested analytes. The participating laboratory shall perform measurements, and report results according to its technical capacity.

As part of the measurement result, the combined standard uncertainty at coverage factor $k=1$ should be reported, taking into account all known uncertainty components.

In case of radioanalytical results, these shall include the applicable contributors listed:

- uncertainty of the calibration source (specified on the certificate);
- uncertainty of the efficiency calibration (due to the efficiency curve fitting and transfer);
- uncertainty of the peak area;
- uncertainty of any applied corrections (self-attenuation, true-coincidence correction, etc.);
- uncertainty coming from the long-term stability of the gamma-ray spectrometer (by control chart if applicable);
- uncertainty from the background subtraction;
- uncertainty from the reproducibility of the geometry (positioning);
- uncertainty of the dry content determination (powder samples); and
- uncertainty due to the sample preparation and chemical recovery (for radiochemistry methods).

In case of ICP-MS results for sample 2 water, and application of a bottom-up approach, the combined uncertainty shall consider all relevant contributors, including (as applicable), but not limited to:

- uncertainty due to measurement precision
- uncertainty of the calibration standard(s) used
- uncertainty of the calibration approach
- uncertainty due to sample preparation, e.g., dilutions
- uncertainty due to correction for blank and any other corrections

Technical instruction for reporting forms

The empty report form can be downloaded from the “<https://analytical-reference-materials.iaea.org/proficiencytests>” page at the “Reference Products for Environment and Trade” webpage.

- radionuclide activity concentrations in sample 1 drinking water

- uranium isotope and total uranium mass fractions in sample 2 natural mineral water
- radionuclide activity concentrations in sample 2 natural mineral water
- radionuclide activity concentrations in sample 3 soil
- radionuclide activity concentrations in sample 4 vegetation
- radionuclide activity in sample 5 simulated contaminated surface sample

In addition to reporting forms, participants are requested to complete a questionnaire on their analytical methods used.

To submit the reporting form, please ensure that you fill in the "Coverpage" sheet with your Lab Code and the Secure Token Code found inside the sample kit box. Both the Lab Code and the Secure Token Code must be filled in and match for your results to be considered authentic.

Please send your result reporting form to the proficiency test contact point at: Proficiency-Tests.Contact-Point@iaea.org.

Include the following information in the subject line of your email: TERM: "PT report IAEA-TERC-2025-01" + your Lab Code

(Example: "PT report IAEA-TERC-2025-01, lab code 901" if your Lab Code is 901)

Note on collusion

Collusion between participants conflicts with professional conduct and negates the benefits of proficiency testing. Any falsification of results to be reported is therefore strongly discouraged in the participants' interest.

5. Evaluation of the reported results

The results of this PT exercise will be evaluated with a commitment to confidentiality of the identity of participants. Every registered participant was assigned a laboratory code, which is only known to the PT provider and the individual participant. The laboratory code will be used to label reported results in the summary report. Without the agreement of the participant, the IAEA will not disclose their performance to a third party.

The reported results will be evaluated according to current IAEA procedures for proficiency testing. Reported results will be compared to a target value. Criteria to evaluate each participant's performance will take into consideration the relative bias of reported values (trueness) as well as the reported uncertainty of measurement (precision).

Target values are assigned independently for most parameters, either by formulation or characterization measurements. In case of results for gross-alpha and gross-beta activity concentrations, a consensus value from reported results will be used.

5.1. Evaluation of parameters with independently assigned target values

Each reported result will be evaluated according to the following steps:

1. The relative bias between the reported and the target value (the best estimate of the true value) is expressed by the following equation:

$$Bias_{relative} = \frac{x_{reported} - x_{target}}{x_{target}} \times 100\%$$

Where:

x_{target} is the target value,

$x_{reported}$ is the value reported by the participant.

2. The relative bias will be compared to the Maximum Acceptable Relative Bias (MARB) which is determined for each analyte, considering the radioanalytical methods, the level of radioactivity and the complexity of the analysis.

If $Bias_{relative} \leq MARB$ value, the result will be "Accepted" for trueness.

3. Based on fit-for-purpose and good laboratory practice principles, the expanded relative uncertainty should cover the relative bias.
4. The P statistic is calculated using the below equation:

$$P = \sqrt{\left(\frac{u_{target}}{x_{target}}\right)^2 + \left(\frac{u_{reported}}{x_{reported}}\right)^2} \times 100\%$$

Where:

u_{target} is the uncertainty of the target value,

$u_{reported}$ is the uncertainty reported by the participant.

5. The P statistic is compared to the relative bias of the reported value:

$$Bias_{relative} \leq k \times P$$

k is the coverage factor, 2.58 corresponding to 99% confidence level.

When the above criterion is fulfilled, the reported result is not significantly different from the target value considering the uncertainties associated with both values. The reported uncertainty of measurement is large enough to cover the bias of the method.

6. The P statistic is also compared to the MARB:

$$P \leq MARB$$

When this criterion is fulfilled, the measurement uncertainty is not overestimated and fit-for-purpose in relation to the MARB criterion of this PT exercise.

7. When both criteria related to the measurement uncertainty are fulfilled, the reported result is rated "accepted (A)" for precision (measurement uncertainty). The result is rated "Not Accepted (N)" for precision if either of the two conditions are not fulfilled. The final score is assigned according to the detailed evaluation described above. The possible scores are listed below:

- "Accepted (A)" when both, trueness and precision were rated "Accepted"
- "Not Accepted (N)" when the trueness rating is "Not Accepted"
- "Warning (W)" when the trueness rating is "Accepted" but the precision rating is "Not Accepted"

In this evaluation method the MARB is a "hard" criterion to emphasize the importance of the measurement trueness, while a "Warning" score indicates either an under- or overestimated uncertainty. According to the above, a "Warning" score means that the result is good in terms of trueness, however the uncertainty estimation should be revised.

8. A z score will be provided in addition to above mentioned parameters due requests by several participants.

$$z = \frac{x_{reported} - x_{target}}{\sigma_{pt}}$$

Where σ_{pt} is the standard deviation for proficiency assessment. The standard deviation for proficiency assessment is determined for each analyte, considering the radioanalytical methods, the level of radioactivity and the complexity of the analysis. In case of independently assigned target values, it is connected to the MARB as follows:

$$\sigma_{PT} = \frac{MARB \cdot x_{target}}{3}$$

5.2. Evaluation of parameters without independently assigned target value ("intercomparison" approach)

When no independently assigned target value is available for a parameter, the consensus from all reported results from PT participants is applied as the target value. Robust statistical methods are applied to derive the target value and the robust standard deviation.

This approach will be applied to gross alpha / beta measurement results.

The evaluation includes calculation of relative bias, z scores and ζ (zeta) scores according to the following steps:

1. The relative bias is calculated as described in Section 5.1.
2. A z score is calculated using the following general equation:

$$z = \frac{x_{reported} - x_{target}}{\sigma_{pt}}$$

Where σ_{pt} is the standard deviation for proficiency assessment.

In this case, the standard deviation for proficiency assessment is the robust standard deviation of the population.

Here, the z score provides information on the deviation of the reported value from the consensus value derived from participants' results, expressed in multiples of the (robust) standard deviation of the population. It should be emphasized that a z score derived this way is a parameter which depends on the population of reported result, i.e. on the performance of participants.

3. The following criteria apply for performance ratings based on obtained z scores:
 - $|z| \leq 2$ accepted (A)
 - $2 < |z| < 3$ questionable (Q)
 - $|z| \geq 3$ not accepted (N)

4. A ζ (zeta) score is calculated using the following general equation:

$$\zeta = \frac{x_{reported} - x_{target}}{\sqrt{(u_{reported})^2 + (u_{target})^2}}$$

The ζ score allows a combined assessment of the reported value and the reported uncertainty of measurement.

5. The following criteria apply for performance ratings based on obtained ζ scores:

- $|\zeta| \leq 2$ accepted (A)
- $2 < |\zeta| < 3$ questionable (Q)
- $|\zeta| \geq 3$ not accepted (N)

Further details on the final evaluation will be provided in the summary evaluation report.